Assignment 4:
Association Rules and Discrimination Analysis

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Assignment: Assignment 4 — Association Rules

1. Introduction

Association Rule Mining extracts interesting relationships between variables in large datasets. An association rule takes the form A → B, meaning that when itemset A is present, itemset B tends to follow. Two key measures evaluate these rules:

```
Support(A -> B)    = Count(A AND B) / Total Transactions
Confidence(A -> B) = Count(A AND B) / Count(A)
```

In the context of credit decisions, association rules can reveal patterns of potential discrimination — where protected attributes such as Gender, combined with socioeconomic attributes such as Income, lead systematically to adverse outcomes such as credit denial. This assignment analyses such patterns using a dataset of 10 credit applications.

2. Dataset

The dataset contains 10 credit application records. Each record includes a protected attribute (Gender), a contextual attribute (Income), a potential proxy variable (Zip_Code), and the final decision (Approved or Denied).

TID	Gender	Income	Zip Code	Decision
1	Female	Low_Income	10451	Denied
2	Male	High_Income	10451	Approved
3	Female	Low_Income	10451	Denied
4	Male	Low_Income	90210	Denied
5	Female	Low_Income	10451	Denied
6	Male	Low_Income	10451	Approved
7	Female	High_Income	90210	Approved
8	Female	Low_Income	10451	Denied
9	Male	High_Income	10451	Approved
10	Female	Low_Income	90210	Approved

Table 1: Credit Application Dataset (N = 10)

Key observations from the raw data: 5 applicants are Female+Low_Income (TIDs 1,3,5,8,10). Of these, 4 were Denied and 1 Approved. There are 5 Denied decisions total (TIDs 1,3,4,5,8).

3. Task 1: Support and Confidence for PD Rule

The Potential Discrimination (PD) rule under analysis is:

Female, Low_Income -> Denied

This rule tests whether being Female and having Low Income systematically leads to credit denial — a potential instance of gender and socioeconomic discrimination.

3.1 Identifying the Relevant Transactions

Antecedent: Female AND Low_Income

Transactions where Gender=Female AND Income=Low_Income:

```
TID 1:  Female, Low_Income, Zip=10451, Denied   <-- matches
TID 3:  Female, Low_Income, Zip=10451, Denied   <-- matches
TID 5:  Female, Low_Income, Zip=10451, Denied   <-- matches
TID 8:  Female, Low_Income, Zip=10451, Denied   <-- matches
TID 10: Female, Low_Income, Zip=90210, Approved <-- matches
Count(Female AND Low_Income) = 5
```

Consequent: Denied

Transactions where Decision=Denied:

```
TID 1, TID 3, TID 4, TID 5, TID 8
Count(Denied) = 5
```

Both Antecedent AND Consequent

Transactions where Female AND Low_Income AND Denied:

```
TID 1:  Female, Low_Income, Denied  <-- matches
TID 3:  Female, Low_Income, Denied  <-- matches
TID 5:  Female, Low_Income, Denied  <-- matches
TID 8:  Female, Low_Income, Denied  <-- matches
Count(Female AND Low_Income AND Denied) = 4
```

Note: TID 10 (Female, Low_Income) was Approved — so it counts in the antecedent but NOT in the combined count.

3.2 Support Calculation

```
Support(Female, Low_Income -> Denied)
= Count(Female AND Low_Income AND Denied) / Total Transactions
= 4 / 10
= 0.4000  (40%)
```

Support = 4/10 = 0.40 (40%)

3.3 Confidence Calculation

```
Confidence(Female, Low_Income -> Denied)
= Count(Female AND Low_Income AND Denied) / Count(Female AND Low_Income)
= 4 / 5
= 0.8000  (80%)
```

Confidence = 4/5 = 0.80 (80%)

The confidence of 80% means that 80% of all Female Low_Income applicants were denied credit. This is a very high rate and warrants further investigation through Extended Lift analysis.

4. Task 2: Extended Lift Calculation

Extended Lift measures whether the inclusion of a protected attribute (Gender=Female) in the rule provides additional discriminatory power beyond what income alone explains. It compares the rule WITH the protected attribute against the same rule WITHOUT it.

```
Extended Lift = Conf(Female, Low_Income -> Denied)
                ______________________________________

                Conf(Low_Income -> Denied)
```

4.1 Non-Discriminatory Reference Rule: Low_Income -> Denied

The reference rule removes the protected attribute (Gender) and keeps only the contextual attribute (Income):

Count(Low_Income)

```
TID 1:  Low_Income  <-- matches
TID 3:  Low_Income  <-- matches
TID 4:  Low_Income  <-- matches
TID 5:  Low_Income  <-- matches
TID 6:  Low_Income  <-- matches
TID 8:  Low_Income  <-- matches
TID 10: Low_Income  <-- matches
Count(Low_Income) = 7
```

Count(Low_Income AND Denied)

```
TID 1:  Low_Income, Denied   <-- matches
TID 3:  Low_Income, Denied   <-- matches
TID 4:  Low_Income, Denied   <-- matches
TID 5:  Low_Income, Denied   <-- matches
TID 8:  Low_Income, Denied   <-- matches
TID 6:  Low_Income, Approved <-- does NOT match
TID 10: Low_Income, Approved <-- does NOT match
Count(Low_Income AND Denied) = 5
```

Confidence of Reference Rule

```
Conf(Low_Income -> Denied) = Count(Low_Income AND Denied) / Count(Low_Income)
                           = 5 / 7
                           = 0.7143  (71.43%)
```

4.2 Extended Lift Calculation

```
Extended Lift = Conf(Female, Low_Income -> Denied)
                ______________________________________
```

```
Conf(Low_Income -> Denied)

= 0.8000 / 0.7143
= 1.1200
```

Extended Lift = 0.8000 / 0.7143 = 1.1200

4.3 Comparison Against Legal Threshold

Metric	Value	Threshold	Exceeds Threshold?
Confidence (PD Rule)	0.8000 (80%)	—	—
Confidence (Reference Rule)	0.7143 (71.43%)	—	—
Extended Lift	1.1200	alpha = 1.5	NO

Table 2: Extended Lift vs Legal Threshold

Extended Lift = 1.1200, which is BELOW the legal discrimination threshold of alpha = 1.5. This means that while Female Low_Income applicants are denied at a higher rate than Low_Income applicants in general (80% vs 71.43%), the added discriminatory effect of Gender is not large enough to formally confirm illegal discrimination by this threshold.

In other words, Income alone explains most of the denial pattern. Gender adds a 12% lift over the income-only baseline — a noteworthy disparity that warrants monitoring, but one that does not formally cross the legal threshold in this small dataset.

5. Task 3: Indirect Discrimination via Proxy Variable Zip_Code='10451'

Indirect discrimination occurs when a seemingly neutral attribute (a proxy) correlates strongly with a protected attribute and produces the same discriminatory effect. Zip_Code is tested as a potential proxy for Female+Low_Income.

5.1 Step 1: Is Zip_Code='10451' a Proxy for Female+Low_Income?

We examine who lives in Zip='10451' versus Zip='90210':

TID	Gender	Income	Decision
1 (Zip=10451)	Female	Low_Income	Denied
2 (Zip=10451)	Male	High_Income	Approved
3 (Zip=10451)	Female	Low_Income	Denied
5 (Zip=10451)	Female	Low_Income	Denied
6 (Zip=10451)	Male	Low_Income	Approved
8 (Zip=10451)	Female	Low_Income	Denied
9 (Zip=10451)	Male	High_Income	Approved

Table 3: Transactions in Zip Code 10451 (7 total)

```
Count(Zip='10451') = 7
Female+Low_Income in Zip='10451' = 4  (TIDs 1, 3, 5, 8)
Total Female+Low_Income = 5  (TIDs 1, 3, 5, 8, 10)

% of Female+Low_Income in Zip='10451' = 4/5 = 80%
```

80% of all Female Low_Income applicants live in Zip='10451'. This is a strong overlap — Zip='10451' disproportionately represents the protected group.

5.2 Step 2: Support and Confidence of Proxy Rule

The proxy rule is: Zip_Code='10451' -> Denied

Count(Zip='10451' AND Denied)

```
TID 1: Zip=10451, Denied  <-- matches
TID 3: Zip=10451, Denied  <-- matches
TID 5: Zip=10451, Denied  <-- matches
TID 8: Zip=10451, Denied  <-- matches
TID 2: Zip=10451, Approved <-- does NOT match
TID 6: Zip=10451, Approved <-- does NOT match
TID 9: Zip=10451, Approved <-- does NOT match
```

```
Count(Zip='10451' AND Denied) = 4

Support(Zip='10451' -> Denied)    = 4 / 10 = 0.4000  (40%)
Confidence(Zip='10451' -> Denied) = 4 / 7  = 0.5714  (57.14%)
```

5.3 Step 3: Extended Lift for Proxy Rule

```
Extended Lift(Zip='10451' -> Denied)
= Conf(Zip='10451' -> Denied) / Conf(Low_Income -> Denied)
= 0.5714 / 0.7143
= 0.8000
```

Extended Lift (Proxy) = 0.5714 / 0.7143 = 0.8000

Extended Lift of 0.80 is BELOW 1.0, meaning Zip='10451' alone is actually a WEAKER predictor of denial than Low_Income alone. The proxy rule does not add discriminatory lift beyond the income baseline.

5.4 Step 4: Correlation Between Proxy and Protected Group

To formally test proxy strength, we compute the correlation between Zip='10451' and Female+Low_Income:

```
P(Female+Low_Income)          = 5/10 = 0.5000
P(Zip='10451')                = 7/10 = 0.7000
P(Female+Low_Income+Zip=10451) = 4/10 = 0.4000

Correlation = P(F+L+Zip) / (P(F+L) x P(Zip))
            = 0.4000 / (0.5000 x 0.7000)
            = 0.4000 / 0.3500
            = 1.1429
```

A correlation of 1.1429 is slightly above 1.0, indicating that Zip='10451' and Female+Low_Income co-occur more often than would be expected by chance alone. However, this is a weak correlation — a value of 1.0 would mean they are statistically independent, and 1.1429 is only marginally above that baseline.

5.5 Summary of Indirect Discrimination Test

Test	Result	Conclusion
% of Female+Low_Income in Zip=10451	80% (4 out of 5)	Strong demographic overlap
Confidence(Zip=10451 -> Denied)	0.5714 (57.14%)	Moderate denial rate
Extended Lift (proxy rule)	0.8000	Below 1.5 threshold — not confirmed

Test	Result	Conclusion
Correlation (proxy vs protected)	1.1429	Weak positive — marginal overlap

Table 4: Indirect Discrimination Test Summary

While 80% of Female Low_Income applicants live in Zip='10451' (suggesting demographic concentration), the proxy rule Zip='10451' -> Denied does not exceed the alpha=1.5 threshold and the correlation is only marginally above independence. Zip='10451' therefore partially proxies the protected group but does not independently drive discrimination beyond what income already explains.

6. Overall Summary and Conclusion

Task	Metric	Value	Threshold	Status
Task 1	Support (PD Rule)	0.4000 (40%)	—	Calculated
Task 1	Confidence (PD Rule)	0.8000 (80%)	—	Calculated
Task 2	Conf(Low_Income->Denied)	0.7143 (71.43%)	—	Reference
Task 2	Extended Lift	1.1200	alpha=1.5	BELOW threshold
Task 3	Conf(Zip=10451->Denied)	0.5714 (57.14%)	—	Calculated
Task 3	Extended Lift (proxy)	0.8000	alpha=1.5	BELOW threshold
Task 3	Correlation (proxy)	1.1429	>1.0	Weak proxy

Table 5: Complete Results Summary

Task 1 found that the PD rule Female, Low_Income -> Denied has a Support of 40% and Confidence of 80%, meaning 4 out of 5 Female Low_Income applicants were denied credit. This is a high denial rate that raises concern about potential discrimination.

Task 2 calculated the Extended Lift as 1.1200 — below the legal threshold of alpha=1.5. Income alone explains most of the denial pattern (71.43% confidence), and adding Gender increases this by only 8.57 percentage points. Formal discrimination is not confirmed by Extended Lift, but the pattern warrants ongoing monitoring.

Task 3 found that Zip='10451' partially proxies for Female+Low_Income (80% overlap) but does not independently drive discrimination. The Extended Lift of the proxy rule (0.80) is below 1.5, and the correlation between Zip='10451' and Female+Low_Income (1.1429) is only marginally above statistical independence. Indirect discrimination via Zip Code is not formally confirmed in this dataset.

In conclusion, the analysis reveals a concerning pattern — Female Low_Income applicants face an 80% denial rate — but the Extended Lift measure does not confirm formal illegal discrimination under the alpha=1.5 threshold. With a larger dataset, the statistical power of these tests would increase and the findings might cross the formal threshold.